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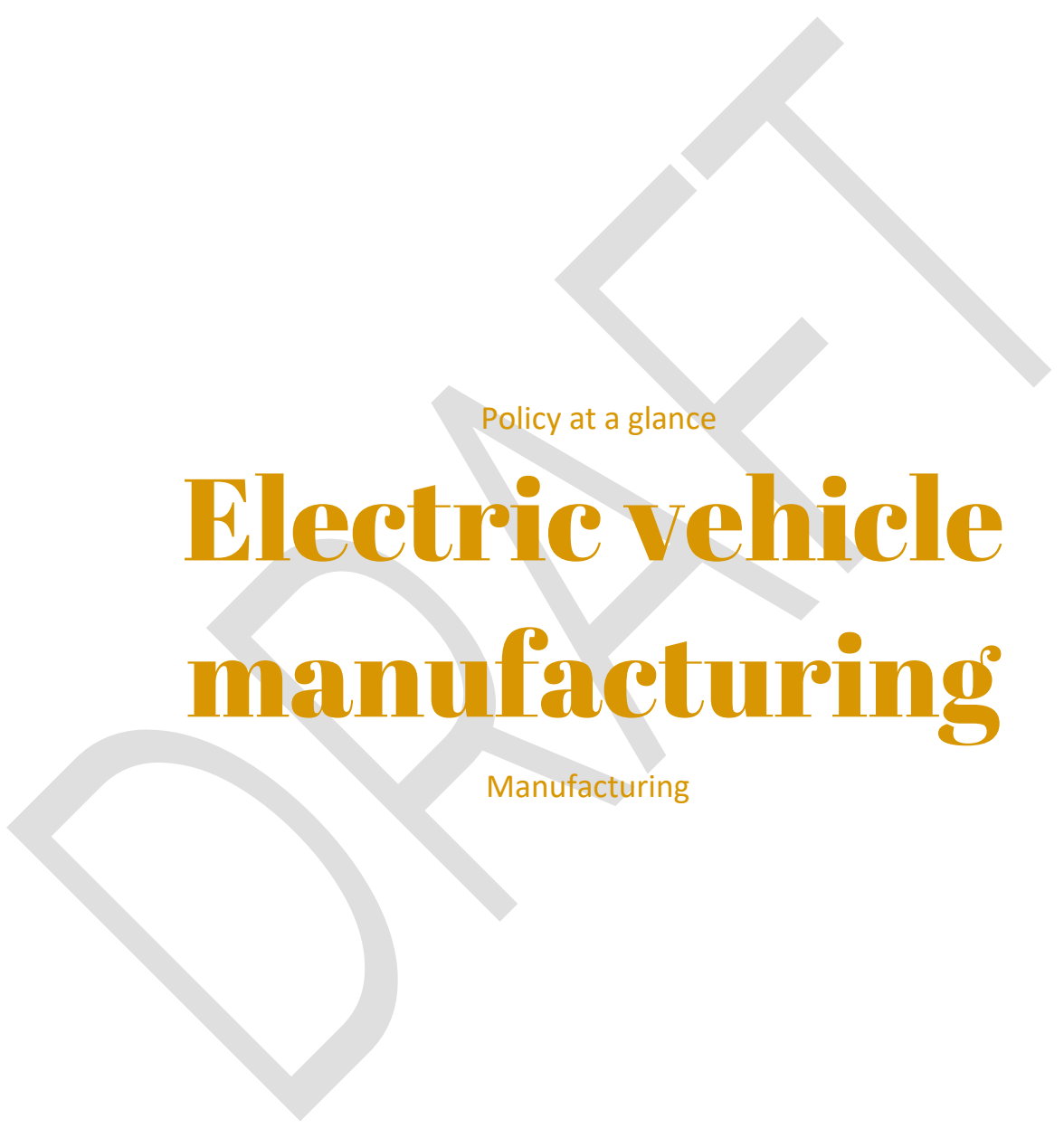
DRAFT

POLICY AT A GLANCE

Electric vehicle

Manufacturing in India

DRAFT



Policy at a glance

Electric vehicle manufacturing

Manufacturing

The '**Policy-at-a-Glance**' provides users with curated information for first-level analysis and inter-state comparisons of policy. The document brings together all topic-relevant national and state-level acts, policies, and schemes in one place. Further, it identifies for a reader what policy features are covered in which document in a simple table.

The first section of this document sets out some 'Context and Background' on the topic; the subsequent sections explain the 'Policy Features' that have been compared and include a 'Glossary' of technical terms used; the last section is the 'Table of Insights' that marks out what 'Policy Feature' has been covered (and on which page) in each of the topic-relevant documents

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1. Context

The automobile industry — one of India's earliest manufacturing success stories — is on the cusp of experiencing one of its largest-ever disruptions with the impending shift away from fossil fuels to alternative modes of propulsion. Electric drivetrain has emerged as a leading mode of propulsion in the global automobile industry. However, the widespread adoption of electric vehicles (EVs) is hindered by challenges such as range anxiety, cost of ownership and acquisition, and development of supporting ecosystem, which are monitorable aspects globally.

The central government has adopted a reward-and-punishment approach to facilitate the transition to Electric. Under this approach, it provides subsidies on both demand and supply sides to smoothen and incentivise the transition to EVs and it also introduces regulations that OEMs have to adhere to or are then hit with financial penalties. State governments, too, have outlined measures to promote EV manufacturing in their states.

The primary objective of this analysis is to evaluate and compare the key features of EV policies in different states. By synthesising information on purchase incentives, capital subsidies, infrastructure support, tax exemptions, skill development initiatives and more, this study aims to offer valuable insights into the approaches adopted by different regions to promote EV manufacturing and build a sustainable EV ecosystem.

2. Policy features

This section identifies a common set of policy features that documents referred to in Section 4 (Table of Insights) cover. The features are classified under the following three broad policy aspects:

- A. Demand subsidies
- B. Supply subsidies
- C. Other incentives

A. Demand subsidies

Purchase incentives: Purchase incentives are financial sops offered by Governments to reduce the higher upfront cost of an Electric vehicle compared to its ICE counterparts. These incentives, often based on battery size or vehicle price, make EVs more affordable and appealing to consumers, driving the shift towards eco-friendly transportation. In certain cases, purchase incentives may also include benefits such as relief from road tax and registration fees, providing additional incentives for adopting EVs.

100% waiver of registration and motor vehicle tax, early bird incentives and lucrative purchase subsidy amounting to 15% of ex-showroom price across all categories except for Light Commercial Vehicles(LCV) has helped [Uttar Pradesh](#) emerge as the state with the highest sales of Electric Vehicles(EV's) across India.

B. Supply subsidies

Capital subsidy: A capital subsidy, part of an EV policy, offers a percentage-based financial incentive to entities establishing EV and battery manufacturing facilities. Calculated from the fixed capital investment, this subsidy reduces initial financial burdens, fostering growth and innovation in the EV manufacturing space. By facilitating sustainable transportation infrastructure development, it accelerates the transition to cleaner mobility solutions.

The [Tamil Nadu EV Policy](#) offers potential investors/OEM's a multitude of options for capital subsidy in the form of Reimbursement of State GST or Turnover based subsidy or Capital subsidy for setting up

new/expansion of existing capacities or special subsidy for setting up Advanced chemistry cells which has led to it emerging as a hub for Electric vehicle manufacturing.

Manufacturing tax exemption: Manufacturing tax exemptions are provided to incentivise the production of EVs and related components. For example, Goods and Services Tax (GST) exemptions help reduce financial burdens for manufacturers and encourage innovation and expansion in the electric mobility industry.

Interest rate subvention: Interest rate subvention in an EV policy incentivises manufacturing by reducing loan interest rates and encouraging companies to invest in EV production. This financial support lowers borrowing costs, fostering growth and innovation in the sustainable transportation sector.

The [Tamil Nadu Electric Vehicle policy](#) provides an interest subvention of 5% as a rebate on the rate of interest, on actual term loans taken for the purpose of financing EV project, for a period of 6 years subject to the limits.

Land development incentive: An EV policy includes land development incentives and land allocation strategies to facilitate the creation of essential infrastructure for EVs. These incentives encompass various benefits, such as reduced conversion charges, stamp duty exemptions and streamlined registration processes, which encourage efficient land usage and the establishment of production units and charging stations.

As part of its [Electric Vehicle policy](#), Chhattisgarh has allocated 500-1000 acres of land for developing EV Parks with plug and play internal infrastructure, common facilities and necessary external infrastructure to industrial park will attract manufacturers across the EV eco-system.

Infrastructure concessions and subsidies: Infrastructure concessions and subsidies play a critical role in EV policies. They provide support for utilities such as electricity, railroads, communication services, sewage treatment plants (STP), and water supply. These concessions create a favourable environment for the development of electric mobility infrastructure, enhancing affordability and feasibility for stakeholders.

The [Haryana Electric vehicle policy](#) provides 100% exemption on Electricity Duty for a period of 20 years. To promote water recycling for manufacturing plant, the Haryana Government reimburses 50% of the cost of water treatment plant up to INR 50 lakh for first 5 units in each category (Medium, Large, Mega and Ultra-Mega units).

C. Other incentives

Skill development: In the context of an EV policy, skill development focuses on enhancing workforce capabilities from an employment perspective. Training and educational programmes are implemented to equip individuals with the necessary skills and knowledge required to meet the evolving demands of the EV industry. This initiative aims to create a skilled workforce that contributes to the growth and sustainability of the sector.

The [Chandigarh Electric Vehicle policy](#) envisages creation of short-term courses on electric mobility, EVSE, repair and maintenance, battery manufacturing and maintenance by the Education Department. Polytechnics are offering two-week duration courses on Electric Vehicle awareness for skill upgradation. Further, skill enhancement centres for delivering vocational courses on the EV ecosystem, with an aim to train the ICE mechanics/workforce in repairing and servicing of EVs and charging stations are being developed.

R&D initiatives: Research and development (R&D) initiatives are an integral part of EV policies, emphasising the importance of research and innovation in advancing EV technology. Through dedicated R&D efforts, the policy promotes the exploration of new ideas, technologies and solutions that enhance the efficiency, performance and overall appeal of EVs, driving progress in the industry.

Educational or Research Institutes setting up R&D centres are being provided subsidy at 50% of project cost up to INR 1 crore for developing new electric charging technology for first 5 units and up to INR 5 crore for developing new electric vehicle technology for first 5 units as per selection of best proposals by the state government in the policy period under the [Haryana Electric Vehicle policy](#).

3. Glossary

ICE vehicles: Internal Combustion Engine (ICE) vehicles, relying on fossil fuels like gasoline or diesel for internal combustion, emit exhaust gases contributing to air pollution and greenhouse gas emissions. Despite advancements in fuel efficiency and emission control, the environmental impact has led to a global shift towards cleaner alternatives like electric vehicles. Efforts to transition to greener transportation underscore the ongoing pursuit of a more eco-friendly and sustainable future.

Electric vehicles: Electric vehicles (EVs) represent a transformative category of automobiles powered by stored electricity, either in rechargeable batteries or other energy storage devices. Unlike traditional internal combustion engine vehicles reliant on fossil fuels, EVs utilize electrical energy for propulsion, with Battery Electric Vehicles (BEVs) relying solely on stored electric power and Plug-in Hybrid Electric Vehicles (PHEVs) combining internal combustion engines with rechargeable batteries. This transition to electric mobility is a crucial global step toward creating more sustainable and environmentally friendly transportation alternatives, reducing greenhouse gas emissions by deriving energy from cleaner and renewable sources.

4. Table of insights

 Geography	 Policy/Reference document	 Aspect: Demand subsidy	 Aspect: Supply subsidies					 Other incentives	
		Purchase subsidy	Capital subsidy	Manufacturing tax exemption	Interest subsidy	Land development incentive	Infrastructure concessions and subsidies	Skill development	R&D initiatives
India	FAME 2	✓ (P 10)							
Andaman and Nicobar	Andaman And Nicobar Islands Electric Vehicle Policy, 2022	✓ (P 11)						✓ (P 13)	
Andhra Pradesh	Andhra Pradesh Electric Mobility Policy, 2018		✓ (P 15)	✓ (P 16)			✓ (P 16)	✓ (P 22)	✓ (P 21)
Arunachal Pradesh	Arunachal Pradesh Electric Vehicle Policy-2021	✓ (P 8)							
Assam	Electric Vehicle Policy of Assam, 2021	✓ (P 7)	✓ (P 9)		✓ (P 9)				✓ (P 10)
Bihar	Bihar Electric Vehicle Policy 2019	✓ (P 4)							
Chandigarh	Chandigarh Electric Vehicle Policy, 2022	✓ (P 5)						✓ (P 11)	✓ (P 10)
Chhattisgarh	Chhattisgarh State Electric Vehicle Policy, 2022	✓ (P 15)	✓ (P 17)	✓ (P 16)		✓ (P 21)		✓ (P 21)	✓ (P 21)
Delhi	Delhi Electric Vehicle Policy, 2020	✓ (P 3)						✓ (P 12)	✓ (P 12)
Goa	Goa Electric Mobility Promotion Policy-2021	✓ (P 6)	✓ (P 5)	✓ (P 5)			✓ (P 6)	✓ (P 14)	✓
Gujarat	Gujarat State Electric Vehicle Policy, 2021	✓ (P 5)	✓ (P 6)		✓ (P 6)	✓ (P 6)	✓ (P 6)	✓ (P 6)	✓ (P 6)
Haryana	Haryana Electric Vehicle Policy, 2022	✓ (P 6)	✓ (P 4)	✓		✓ (P 5)	✓ (P 4)	✓ (P 8)	✓ (P 8)
Himachal Pradesh	Himachal Pradesh Electric Vehicle Policy, 2022	✓ (P 28)		✓ (P 30)	✓ (P 30)	✓ (P 30)	✓ (P 30)		

 Geography	 Policy/Reference document	 Aspect: Demand subsidy	 Aspect: Supply subsidies					 Other incentives	
		Purchase subsidy	Capital subsidy	Manufacturing tax exemption	Interest subsidy	Land development incentive	Infrastructure concessions and subsidies	Skill development	R&D initiatives
Jharkhand	Jharkhand Electric Vehicle Policy, 2022	✓ (P 9)	✓ (P 5)		✓ (P 7)		✓ (P 8)		
Karnataka	Karnataka Electric Vehicle and Energy Storage Policy, 2017		✓ (P 17)	✓ (P 20)		✓ (P 18)	✓ (P 18)	✓ (P 15)	✓ (P 14)
Kerala	Kerala Electric Vehicle Policy, 2019	✓ (P 9)						✓ (P 9-10)	✓ (P 9)
Ladakh	Ladakh Electric Vehicle and Allied Infrastructure Policy, 2022	✓ (P 7)						✓ (P 8)	✓ (P 8)
Madhya Pradesh	Madhya Pradesh Electric Vehicle (EV) Policy 2019	✓ (P 14)	✓ (P 31)		✓ (P 31)	✓ (P 31)		✓ (P 32)	✓ (P 31)
Maharashtra	Maharashtra Electric Vehicle Policy, 2021	✓ (P 5)	✓ (P 11)	✓ (P 11)	✓ (P 11)	✓ (P 11)	✓ (P 11)	✓ (P 11)	✓ (P 10)
Manipur	Manipur Electric Mobility Policy, 2022	✓ (P 4)						✓ (P 5)	
Meghalaya	Meghalaya Electric Vehicle Policy, 2021	✓ (P 16)						✓ (P 13)	
Odisha	Odisha Electric Vehicle Policy, 2021	✓ (P 7)	✓ (P 11)	✓ (P 10)				✓ (P 16)	✓ (P 17)
Punjab	Punjab Electric Vehicle Policy, 2022	✓ (P 7)	✓ (P 20)	✓ (P 19)		✓ (P 19)	✓ (P 19)	✓ (P 22)	✓ (P 21)
Rajasthan	Rajasthan Electric Vehicle Policy (REVP), 2022	✓ (P 10)	✓ (P 17)	✓ (P 17)	✓ (P 17)	✓ (P 17)	✓ (P 17)	✓ (P 18)	✓ (P 16)
Tamil Nadu	Tamil Nadu Electric Vehicle Policy 2023	✓ (P 22)	✓ (P 16)	✓ (P 17)	✓ (P 18)	✓ (P 17)	✓ (P 15)	✓ (P 21)	✓ (P 22)
Telangana	Telangana Electric Vehicle and Energy Storage Policy, 2020	✓ (P 10)	✓ (P 13)	✓ (P 13)	✓ (P 13)		✓ (P 13)	✓ (P 13)	✓ (P 14)
Uttar Pradesh	Uttar Pradesh Electric Vehicle Manufacturing and Mobility Policy, 2022	✓ (P 16)	✓ (P 21)			✓ (P 16)		✓ (P 23)	✓ (P 23)

 Geography	 Policy/Reference document	 Aspect: Demand subsidy	 Aspect: Supply subsidies					 Other incentives	
		Purchase subsidy	Capital subsidy	Manufacturing tax exemption	Interest subsidy	Land development incentive	Infrastructure concessions and subsidies	Skill development	R&D initiatives
Uttarakhand	Uttarakhand Electric Vehicle Policy 2019	✓						✓	✓
West Bengal	West Bengal Electric Vehicle Policy 2021							✓ (P 8)	✓ (P 8)

Note 1: Documents linked in the tables include original acts, policies, schemes, programmes and guidelines. It is aimed at highlighting and enabling a comparison of relevant policy features covered in the documents, across the centre and states. While all important documents have been identified, the coverage may not be exhaustive. Further, 'Policy at a glance' is not aimed at analysing nuances of policies.

Note 2: Page numbers specified are as per the PDF pagination opened in a Google Chrome browser. It is possible to encounter differences in the actual page number for some of the documents when opened in a different web browser/application.

5. References

- Ministry of Heavy Industries
- Ministry of Road Transport and Highways



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